

Planar Graph

Introduction

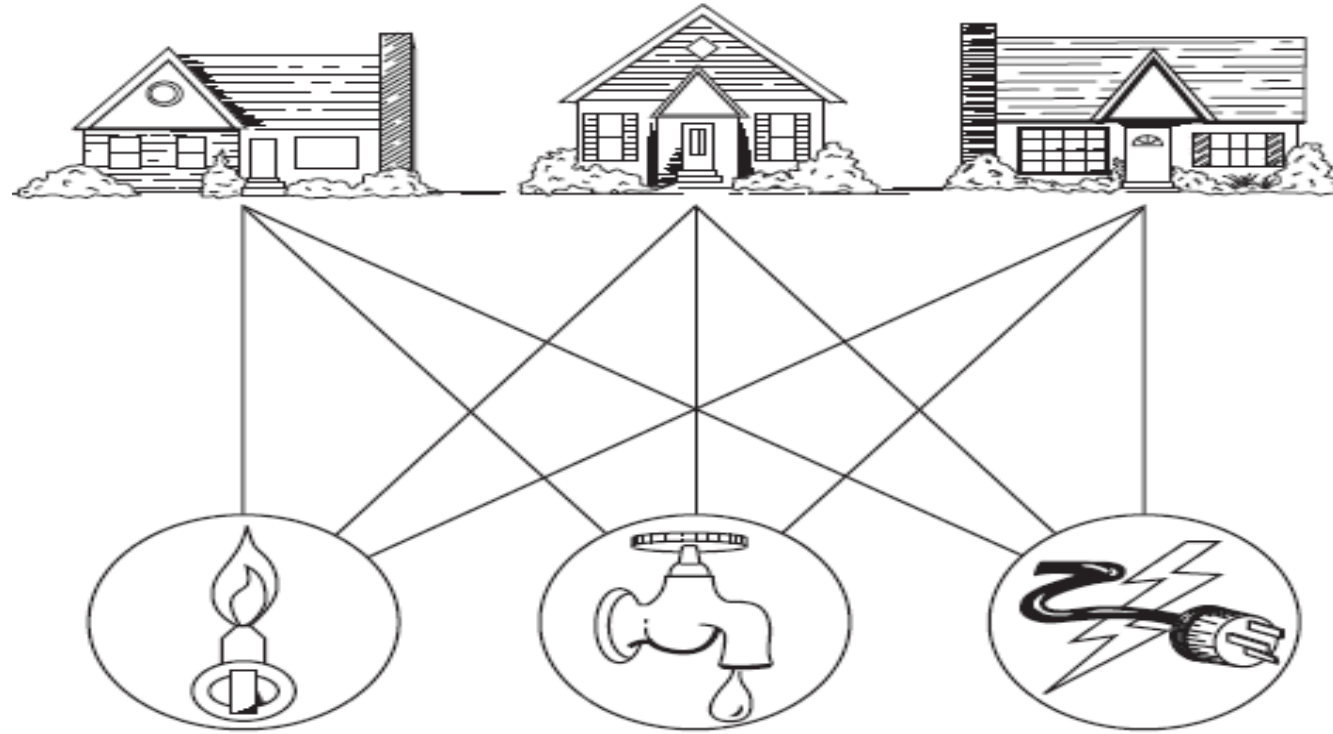


FIGURE 1 Three Houses and Three Utilities.

Planar Graph

- A graph is called *planar* if it can be drawn in the plane without any edges crossing. Such a drawing is called a *planar representation* of the graph where a crossing of edges is the intersection of the lines or arcs.
- A graph may be planar even if it is usually drawn with crossings, because it may be possible to draw it in a different way without crossings.

Example of Planar Graph

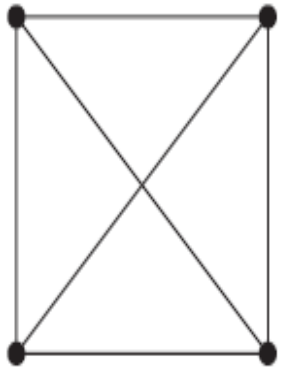


FIGURE 2 The Graph K_4 .

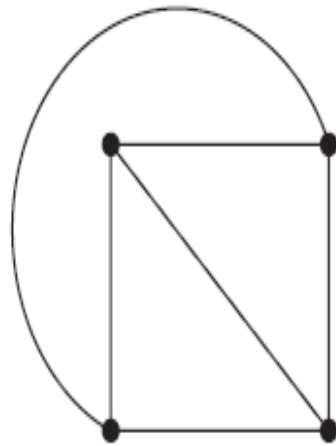


FIGURE 3 K_4 Drawn with No Crossings.

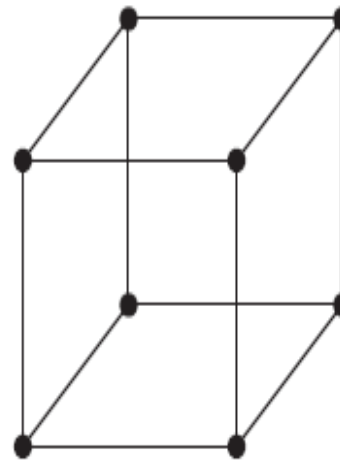


FIGURE 4 The Graph Q_3 .

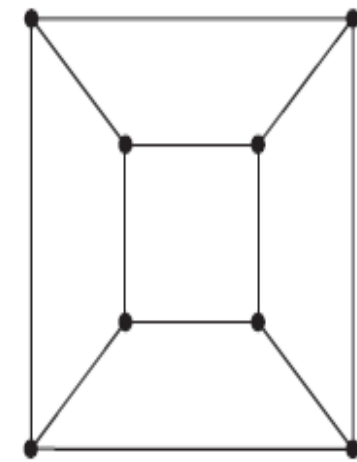
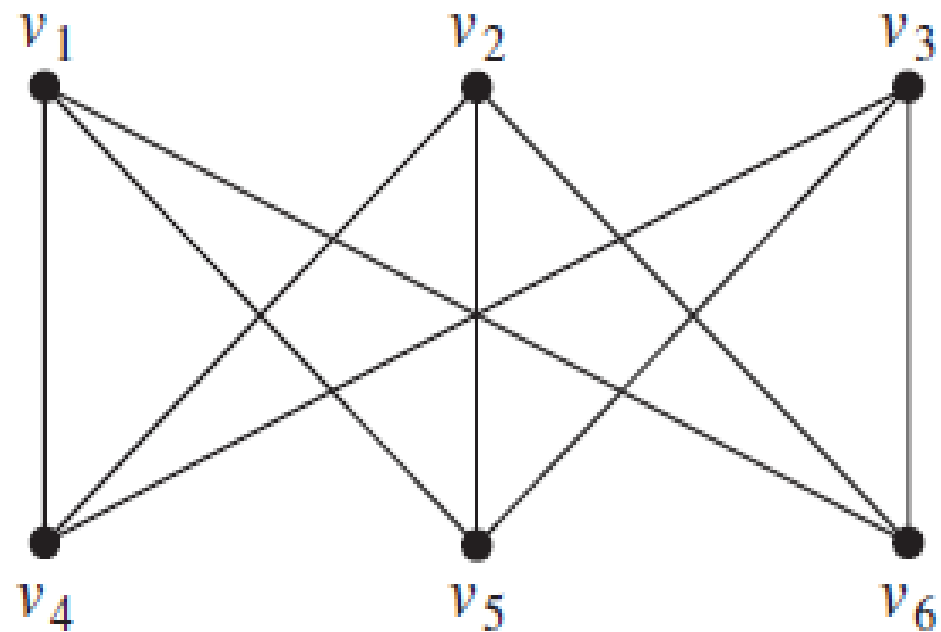


FIGURE 5 A Planar Representation of Q_3 .

Is $K_{3,3}$, shown in Figure, planar?



- In any planar representation of $K_{3,3}$, the vertices v_1 and v_2 must be connected to both v_4 and v_5 . These four edges form a closed curve that splits the plane into two regions, R_1 and R_2 , as shown in Figure 7(a).
- The vertex v_3 is in either R_1 or R_2 . When v_3 is in R_2 , the edges between v_3 and v_4 and between v_3 and v_5 separate R_2 into two sub regions, R_{21} and R_{22} , as shown in Figure 7(b).

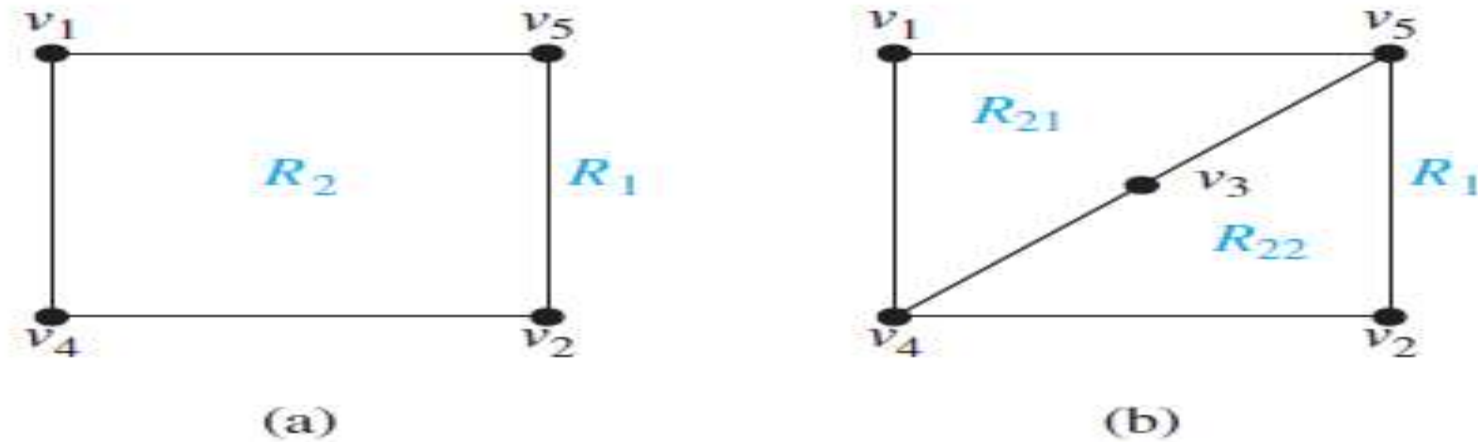


FIGURE 7 Showing that $K_{3,3}$ Is Nonplanar.

- There is no way to place the final vertex v_6 without forcing a crossing.
- If v_6 is in R_1 , then the edge between v_6 and v_3 cannot be drawn without a crossing.
- If v_6 is in R_{21} , then the edge between v_2 and v_6 cannot be drawn without a crossing.
- If v_6 is in R_{22} , then the edge between v_1 and v_6 cannot be drawn without a crossing.

- **EULER'S FORMULA** Let G be a connected planar simple graph with e edges and v vertices. Let r be the number of regions in a planar representation of G . Then $r = e - v + 2$.
- **EXAMPLE:** Suppose that a connected planar simple graph has 20 vertices, each of degree 3. Into how many regions does a representation of this planar graph split the plane?
- This graph has 20 vertices, each of degree 3, so $v = 20$. Because the sum of the degrees of the vertices, $3v = 3 \cdot 20 = 60$, is equal to twice the number of edges, $2e$, we have $2e = 60$, or $e = 30$.
- From Euler's formula, the number of regions is $r = e - v + 2 = 30 - 20 + 2 = 12$.

- **COROLLARY 1** If G is a connected planar simple graph with e edges and v vertices, where $v \geq 3$, then $e \leq 3v - 6$.
- **COROLLARY 2** If G is a connected planar simple graph, then G has a vertex of degree not exceeding five.
- **COROLLARY 3** If a connected planar simple graph has e edges and v vertices with $v \geq 3$ and no circuits of length three, then $e \leq 2v - 4$.
- **EXAMPLE 6** Use Corollary 3 to show that $K_{3,3}$ is nonplanar.

Solution: Because $K_{3,3}$ has no circuits of length three (this is easy to see because it is bipartite), Corollary 3 can be used. $K_{3,3}$ has six vertices and nine edges. Because $e = 9$ and $2v - 4 = 8$, Corollary 3 shows that $K_{3,3}$ is nonplanar.